KENNETH BELL

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EDUCATION

UC San Diego La Jolla, CA
B.S. Electrical Engineering *Expected Grad. – Spring 2029*

- GPA: **3.83** | **Provost Honors** *(Fall 25, Winter 26, Spring 26)*
- Relevant Coursework: **Analog Design** (**A**), **Circuits and Systems** (**A+**), Physics E&M (**A**)
  In Progress: Linear Systems Fundamentals, Engineering Probability and Statistics

PROJECTS

PPG Vitals Monitor — Biosensor System June 2026 – Present
Independent Project

- Designed and simulated a multi-stage analog front end for a photoplethysmography sensor in **LTspice**: photodiode → transimpedance amplifier → passive high/low-pass filters → inverting amplifier
- Captured the schematic and laid out a **2-layer PCB** in **KiCad**; hand-soldered and debugged the assembled board
- Developed embedded **C++** firmware for a Teensy 4.0 using PlatformIO and Arduino to sample the analog sensor and stream measurements for test and analysis
- Built a Python test tool with NumPy and Matplotlib to display the live waveform and **FFT spectrum**, extract heart rate, and verify that signal energy fell within the designed filter passband
- Modeled and 3D-printed a FreeCAD enclosure to align the LED and photodiode across a finger

Hazard Detector for Mobility Devices Spring 2026
IEEE at UCSD

- Implemented a **Python** computer-vision pipeline using SAM2 segmentation to identify a mobility device's traversable path and flag hazards within its boundary
- Collaborated with teammates through **Git** on a shared codebase and integrated detection output into a full-stack testing dashboard with logging and visual feedback

TECHNICAL SKILLS

Hardware	PCB Layout & Assembly, Soldering, Oscilloscope, Function Generator, DMM
Embedded	Teensy 4.0, PlatformIO, Arduino
Tools	KiCad, LTspice, ngspice, FreeCAD, Git, Linux
Languages	C++, Python (NumPy, Matplotlib), MATLAB, Rust, Go, Java